

GFS128 • RME128

相位解缠数据集图册

Dataset Atlas for Phase Unwrapping

数据协议

分辨率	128 × 128
训练集	每类 5,000 张; SNR ∈ {0, 5, 10, 20, 30, 60} dB
测试集	每类每个 SNR 1,000 张; 0/5/10/20/30 dB, 共 5,000 张
字段	φ : 干净展开相位; ψ : 含噪包裹相位; scene_id: 场景编号
生成随机种子	20240913

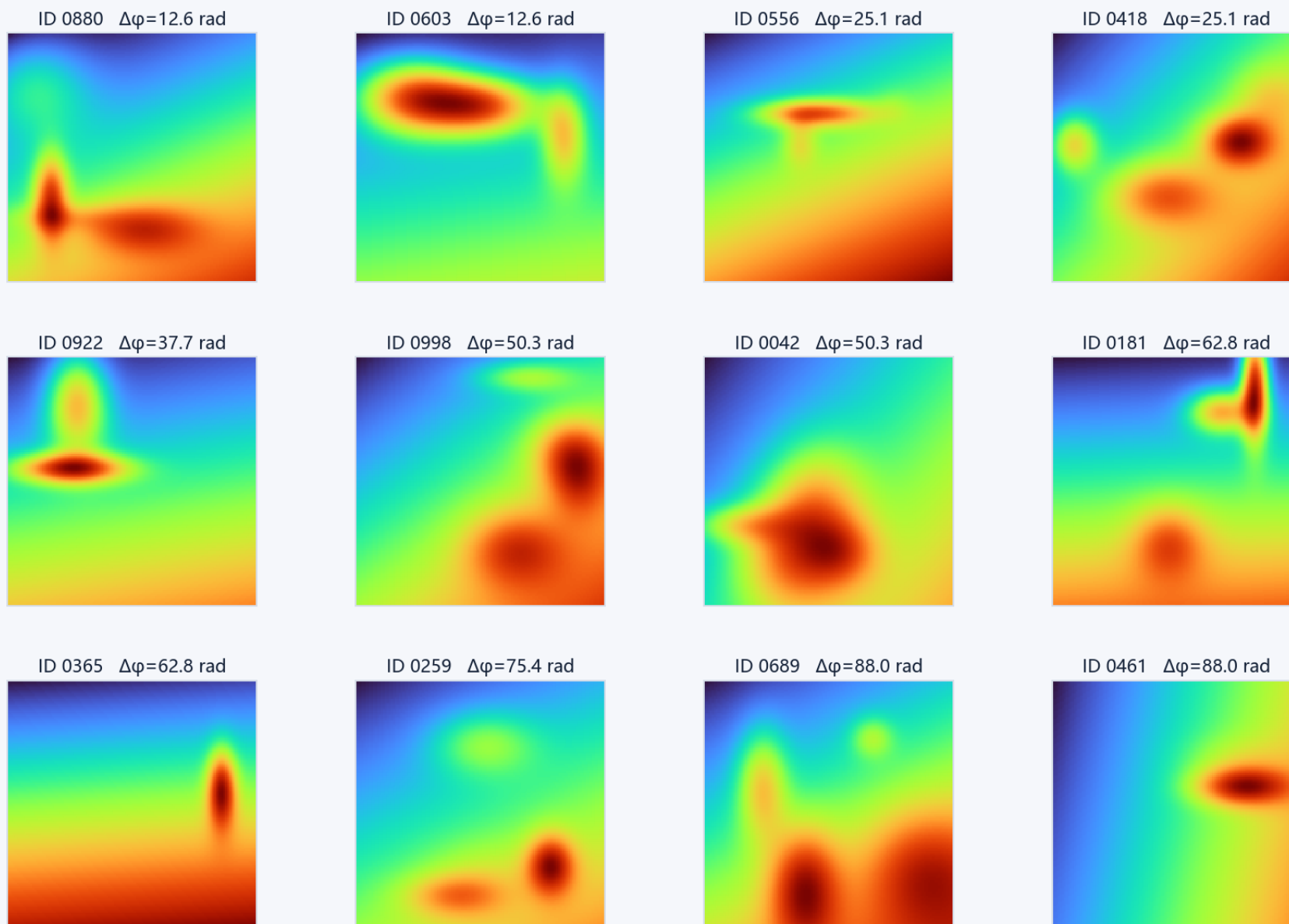
代表样本

GFS128: [880, 603, 556, 418, 922, 998, 42, 181, 365, 259, 689, 461]

RME128: [246, 769, 704, 615, 866, 204, 792, 603, 14, 871, 810, 475]

GFS128 · 干净展开相位形态

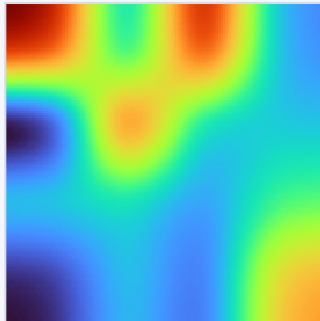
按相位跨度分位数选取 12 个样本，每幅图独立色标范围，以突出空间结构。



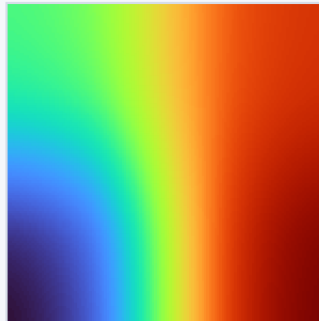
RME128 · 干净展开相位形态

按相位跨度分位数选取 12 个样本，每幅图独立色标范围，以突出空间结构。

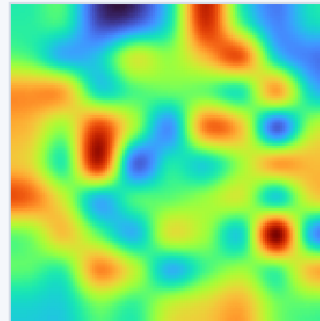
ID 0246 $\Delta\varphi=12.6$ rad



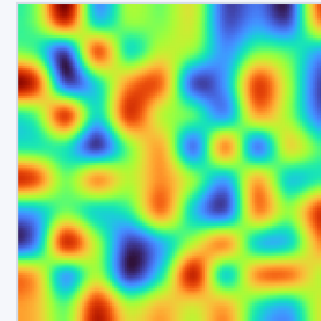
ID 0769 $\Delta\varphi=12.6$ rad



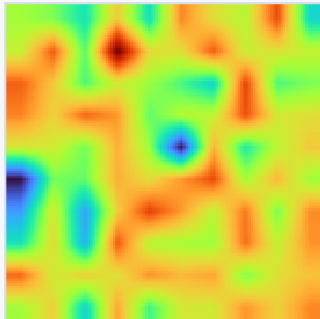
ID 0704 $\Delta\varphi=25.1$ rad



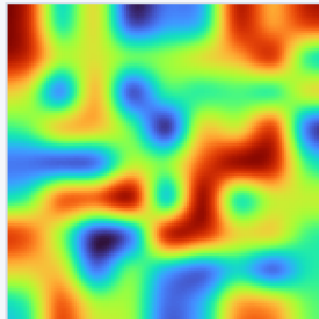
ID 0615 $\Delta\varphi=37.7$ rad



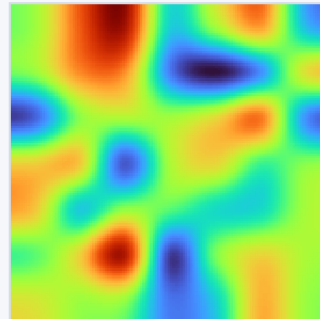
ID 0866 $\Delta\varphi=37.7$ rad



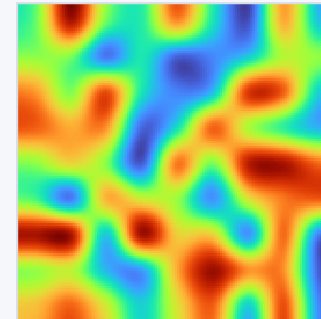
ID 0204 $\Delta\varphi=50.3$ rad



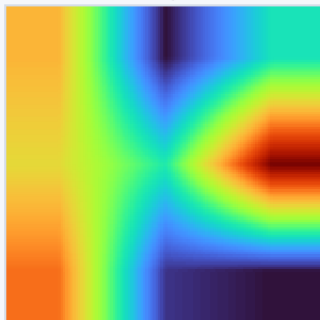
ID 0792 $\Delta\varphi=50.3$ rad



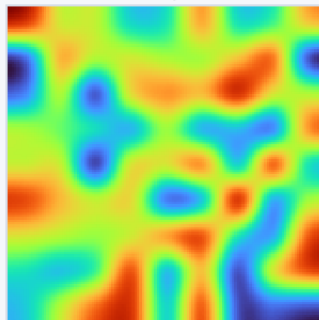
ID 0603 $\Delta\varphi=62.8$ rad



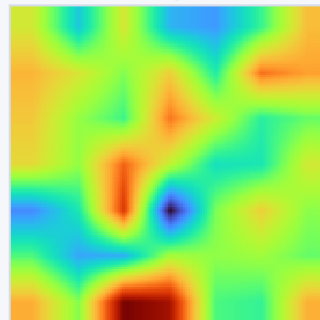
ID 0014 $\Delta\varphi=75.4$ rad



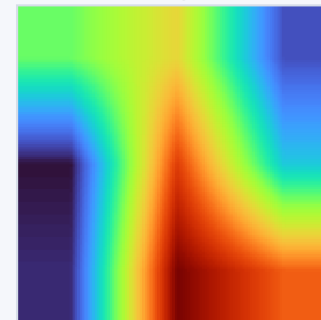
ID 0871 $\Delta\varphi=75.4$ rad



ID 0810 $\Delta\varphi=88.0$ rad

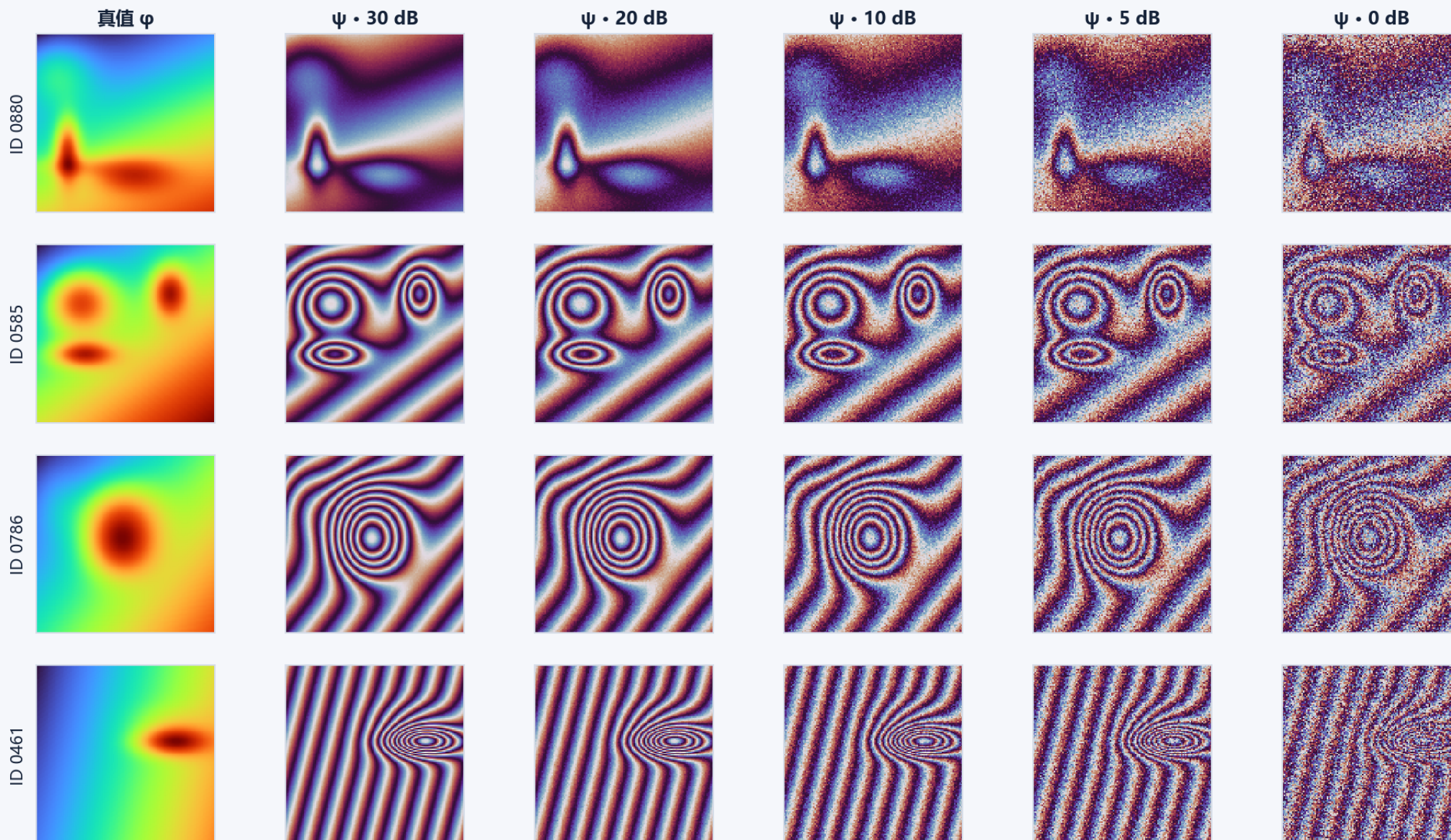


ID 0475 $\Delta\varphi=88.0$ rad



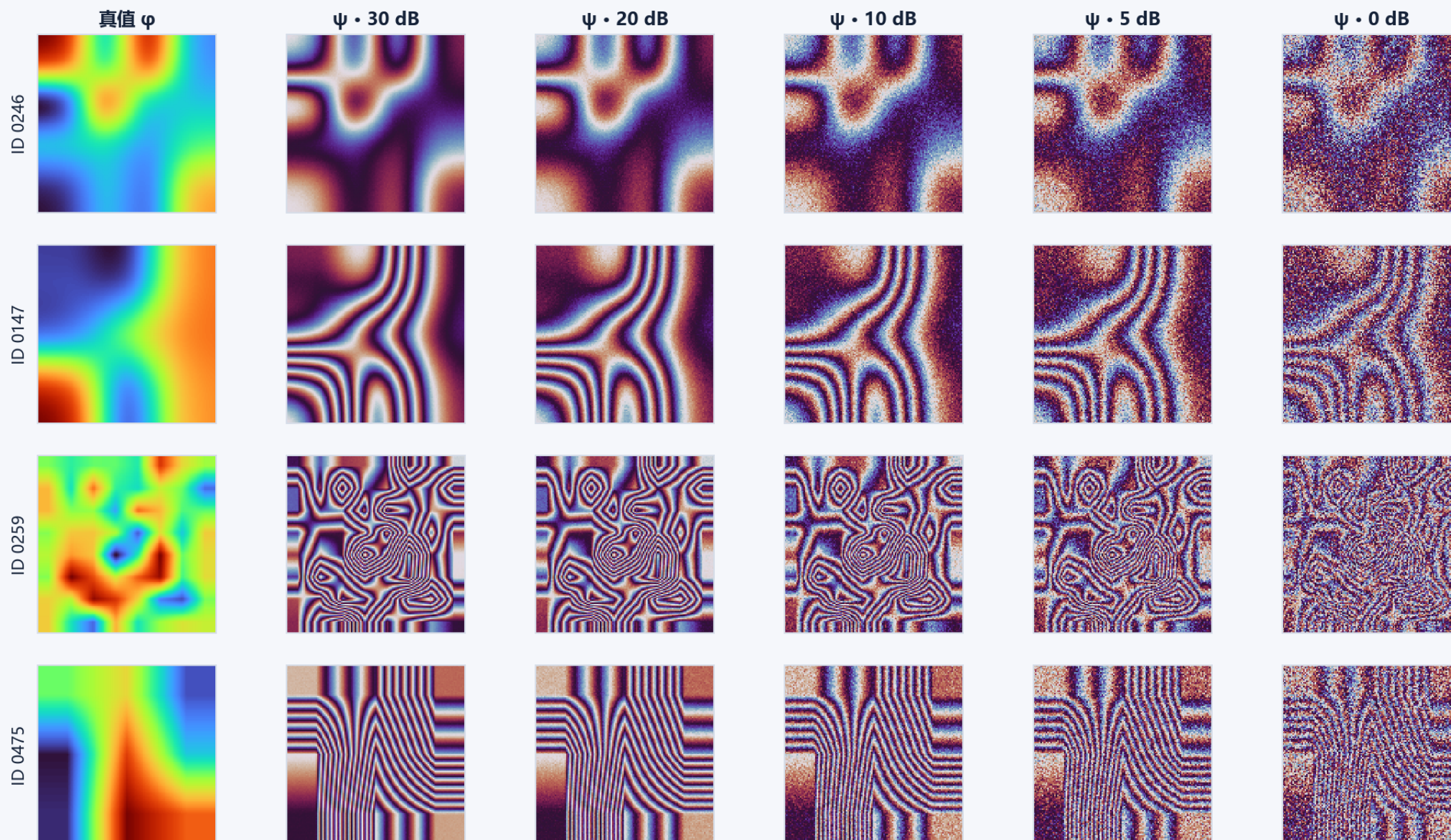
GFS128 · 同一场景在不同噪声等级下的观测

第一列为展开真值 φ ；其余各列为含噪包裹相位 ψ ，统一使用 $[-\pi, \pi]$ 色标。



RME128 · 同一场景在不同噪声等级下的观测

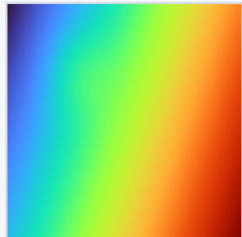
第一列为展开真值 φ ；其余各列为含噪包裹相位 ψ ，统一使用 $[-\pi, \pi]$ 色标。



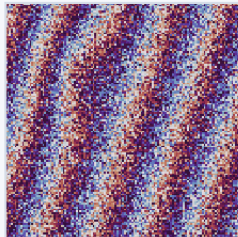
GFS128 · 训练集样本对

每组左侧为展开真值，右侧为训练时输入的含噪包裹相位；覆盖训练集全部 SNR 档位。

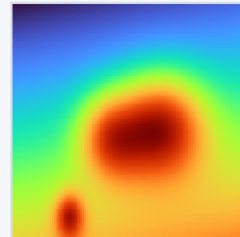
ID 0961 · 真值



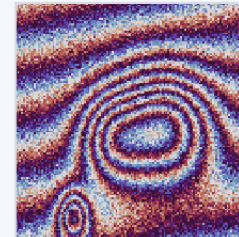
输入 · 0 dB



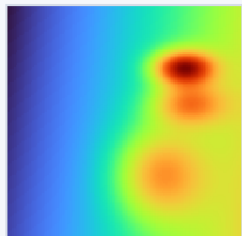
ID 3637 · 真值



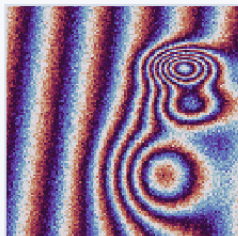
输入 · 5 dB



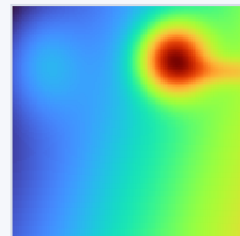
ID 0237 · 真值



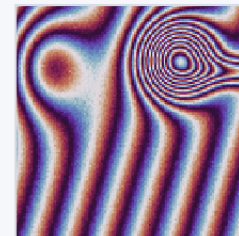
输入 · 10 dB



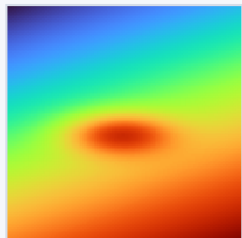
ID 1294 · 真值



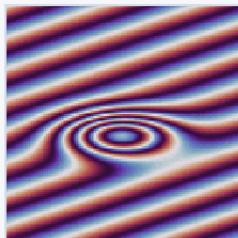
输入 · 20 dB



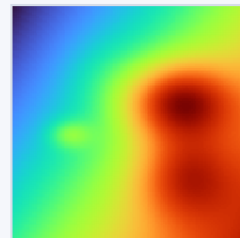
ID 0117 · 真值



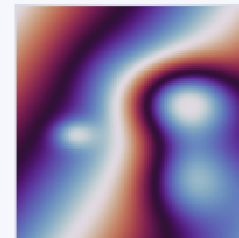
输入 · 30 dB



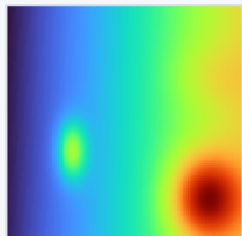
ID 1656 · 真值



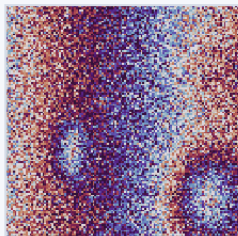
输入 · 60 dB



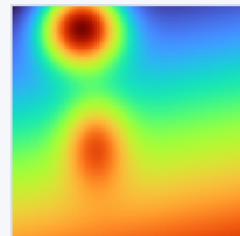
ID 0832 · 真值



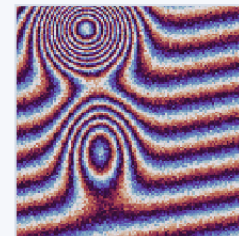
输入 · 0 dB



ID 1170 · 真值



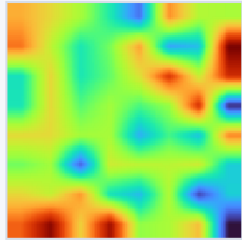
输入 · 10 dB



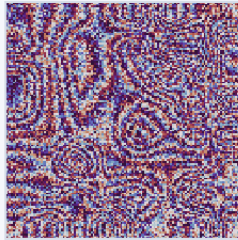
RME128 · 训练集样本对

每组左侧为展开真值，右侧为训练时输入的含噪包裹相位；覆盖训练集全部 SNR 档位。

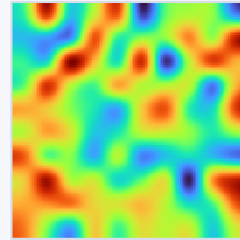
ID 4261 · 真值



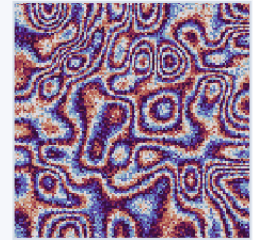
输入 · 0 dB



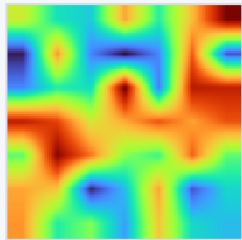
ID 2865 · 真值



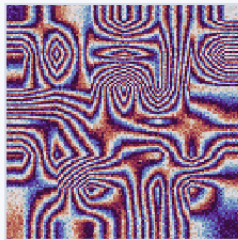
输入 · 5 dB



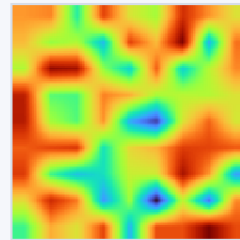
ID 2666 · 真值



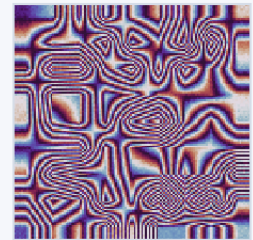
输入 · 10 dB



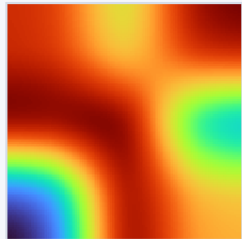
ID 0794 · 真值



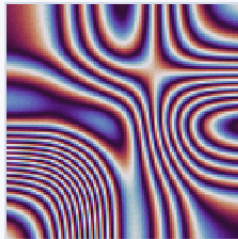
输入 · 20 dB



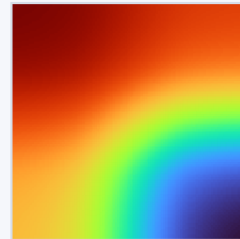
ID 4063 · 真值



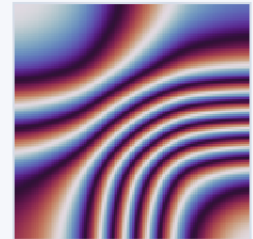
输入 · 30 dB



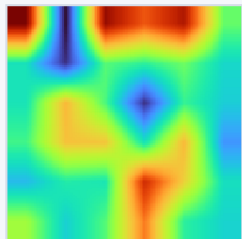
ID 0772 · 真值



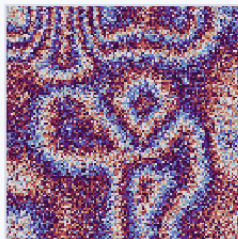
输入 · 60 dB



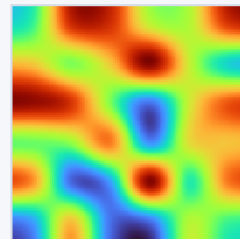
ID 4421 · 真值



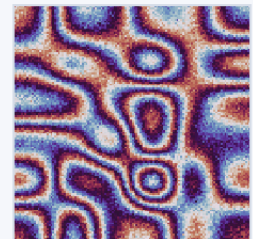
输入 · 0 dB



ID 2420 · 真值



输入 · 10 dB



GFS128 与 RME128 · 数据分布概览

统计基于 1,000 个固定测试场景；噪声误差按圆周距离计算。

